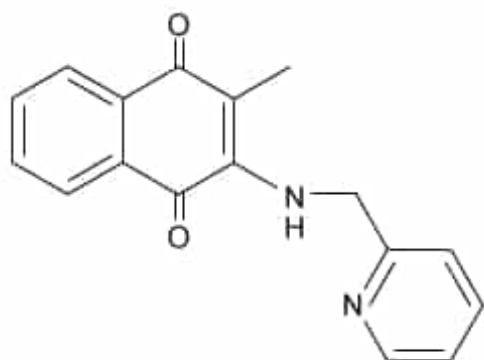


A solution of Compound **1**, shown below, absorbs light maximally at 448 nm in the absence of copper(II) ions but shifts to a 623 nm absorption maximum upon the addition of Cu^{2+} . Which of the following best describes this process?



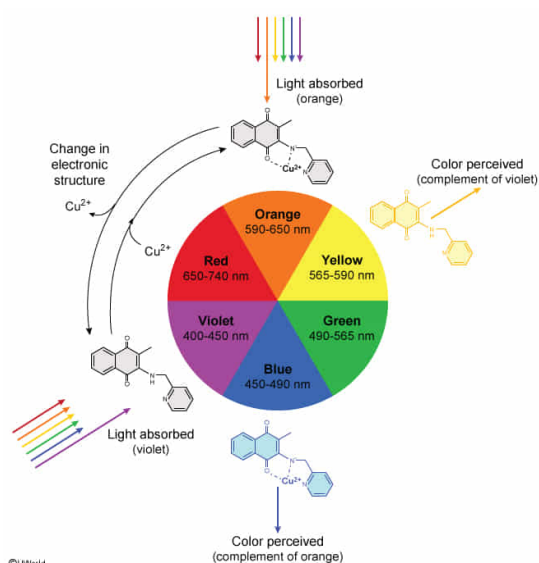
Compound **1**

- ✓ ☒ A. Changes in electronic structure cause the solution to change from yellow to blue. [38%]
- ☐ B. Changes in vibrational modes cause the solution to change from green to yellow. [17%]
- ☐ C. Changes in the mass-to-charge ratio (m/z) cause the solution to change from violet to orange. [40%]
- ☐ D. Changes in nuclear spin cause the solution to change from colorless to violet. [3%]

Correct

37% answered correctly

Explanation:



In **highly conjugated systems**, the difference in energy between the ground and excited states of the electrons is equal to the energy of a particular wavelength of visible light. Photons of this wavelength are absorbed by the molecule, causing an electron to enter the excited state. The remaining wavelengths are either **reflected or transmitted** and are ultimately perceived by the eye as various colors.

In general, the **perceived color** of a substance is **complementary** to the color of the wavelength that is maximally absorbed by that substance. For example, a molecule that absorbs blue light will appear orange and vice versa. When the basic colors of the visible spectrum (red, orange, yellow, green, blue, and violet, or ROYGBV) are arranged in a wheel with red and violet next to each other, a color's complement is the color directly across from it on the wheel.

According to the given information, Compound **1** **absorbs violet light** (448 nm), so it **appears yellow** (complement of violet). After Cu^{2+} is added, the absorption maximum changes to 623 nm and the compound **absorbs orange light**, causing it to **appear blue** (complement of orange). Therefore, upon the addition of Cu^{2+} , Compound **1** must undergo a change in electronic structure that causes the solution to change from **yellow to blue**.

(Choice B) Vibrational modes are excited by **infrared light**, which is not visible to the eye. They do not affect the color of a solution.

(Choice C) The mass-to-charge ratio (m/z) is measured for ions generated from samples analyzed by **mass spectrometry** and pertains to the molecular mass of the ions, which does not affect color of the sample. Compound **1** *absorbs* violet light initially and then *absorbs* orange light when Cu^{2+} is added; the color perceived is *not* the color absorbed, but rather the *complement* of the color absorbed.

(Choice D) Nuclear spin is probed by radio frequencies in **nuclear magnetic resonance (NMR) spectroscopy** and does not affect color.

Educational objective:

The color of a substance is determined by the wavelengths of light it absorbs, which is determined by the electronic structure of the molecule. In general, a substance appears to be the color that is complementary to the color of light that is maximally absorbed.

Time spent:0 sec

Subject:
Organic Chemistry

QID:401442

System:

Separation Techniques, Spectroscopy, and Analytical Methods